

Aufgabe 1: (25 Punkte)

1.

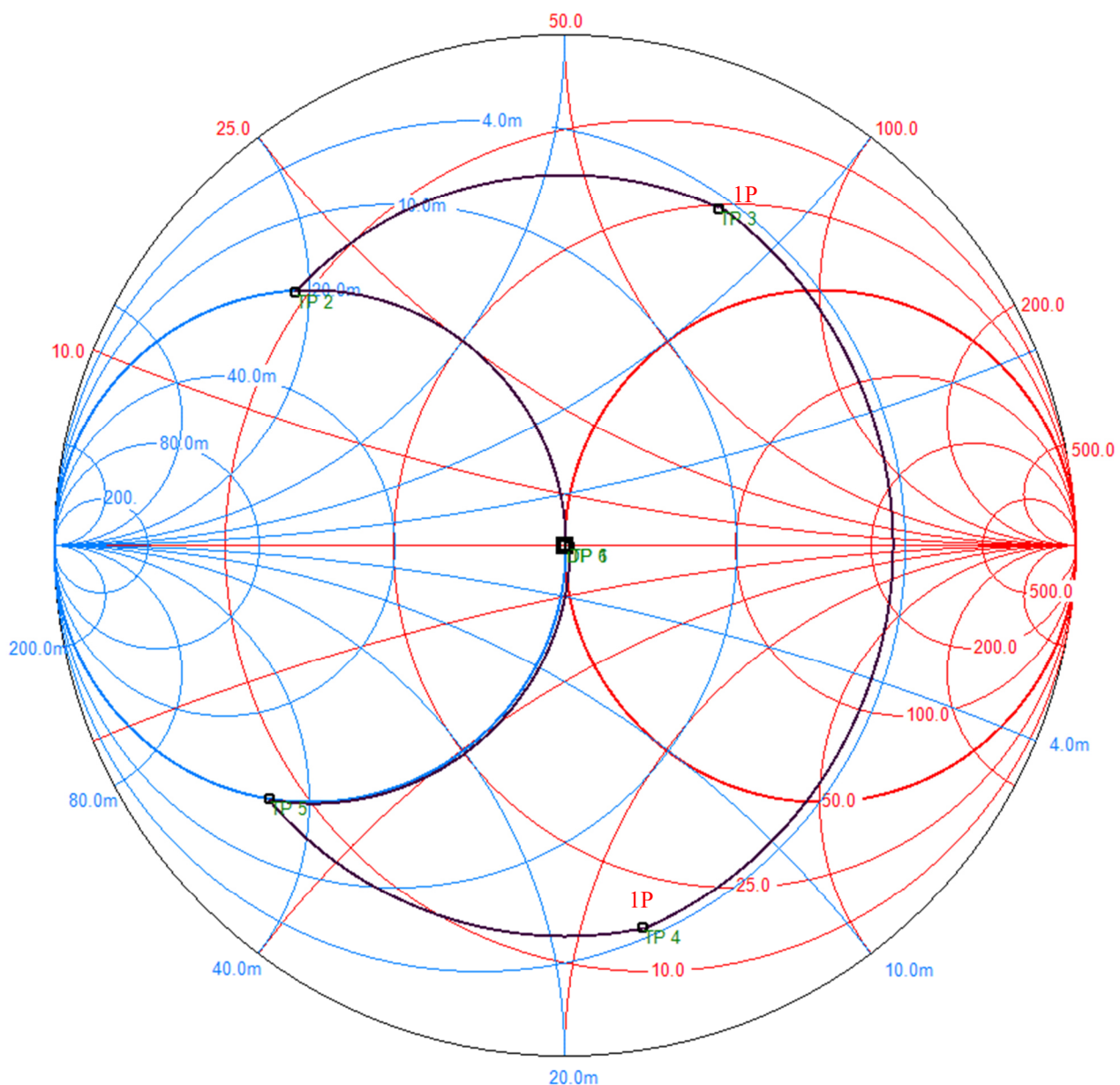
$$X_L = 2\pi \cdot f \cdot L = 2\pi \cdot 1,5 \cdot 10^9 \frac{1}{s} \cdot 2,5 \cdot 10^{-9} \frac{Vs}{A} = 23,56\Omega$$

$$X_C = -\frac{1}{2\pi \cdot f \cdot C} = -\frac{1}{2\pi \cdot 1,5 \cdot 10^9 \frac{1}{s} \cdot 3,0 \cdot 10^{-12} \frac{As}{V}} = -35,37\Omega; \text{ normiert: } \frac{X_C}{Z_L} = \frac{-35,37\Omega}{50\Omega} = 0,7074$$

$$\underline{Z}_1 = \frac{R \cdot jX_L}{R + jX_L} = \frac{50\Omega \cdot j23,56\Omega}{50\Omega + j23,56\Omega} = (9,1 + j19,3)\Omega; \text{ normiert: } \frac{\underline{Z}_1}{Z_L} = \frac{(9,1 + j19,3)\Omega}{50\Omega} = 0,1817 + j0,3856$$

→ ins Smith-Chart eintragen oder: Einzelwerte R und L in Smith-Chart eintragen.

$$\lambda = \frac{c_0}{\sqrt{\epsilon_r} \cdot f} = \frac{3 \cdot 10^8 \frac{m}{s}}{\sqrt{2,0} \cdot 1,5 \cdot 10^9 \frac{1}{s}} = 141,4\text{mm} \quad \rightarrow \quad \frac{l_1}{\lambda} = \frac{14\text{mm}}{141,4\text{mm}} = 0,099$$

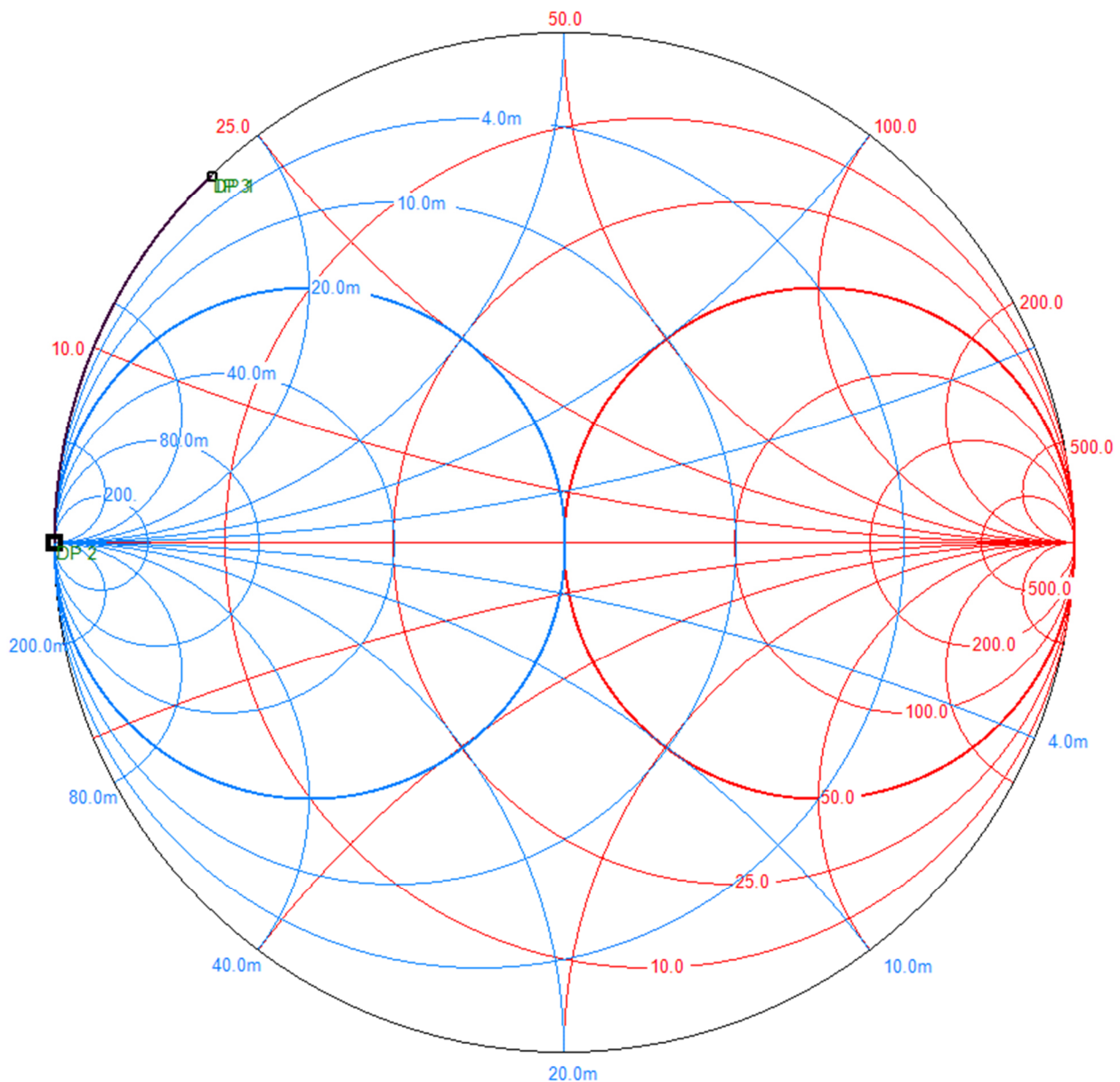


aus Smith-Chart abgelesen:

$$\frac{l_2}{\lambda} = 0,4432 - 0,3586 = 0,0846 \quad \underline{\underline{l_2 = 0,0846 \cdot 141,4\text{mm} = 12,0\text{mm}}}$$

$$B_{L_2} \cdot Z_L = -2,347 \rightarrow |\underline{Z}| = \omega L_2 = \frac{Z_L}{2,347} \quad \underline{\underline{L_2 = \frac{Z_L}{\omega \cdot 2,347} = \frac{50 \frac{\text{V}}{\text{A}}}{2\pi \cdot 1,5 \cdot 10^9 \frac{1}{\text{s}} \cdot 2,347} = 2,3\text{nH}}}$$

2. kurzgeschlossene Stichleitung für $\underline{Z} = j21,3\Omega$:



aus Smith-Chart abgelesen:

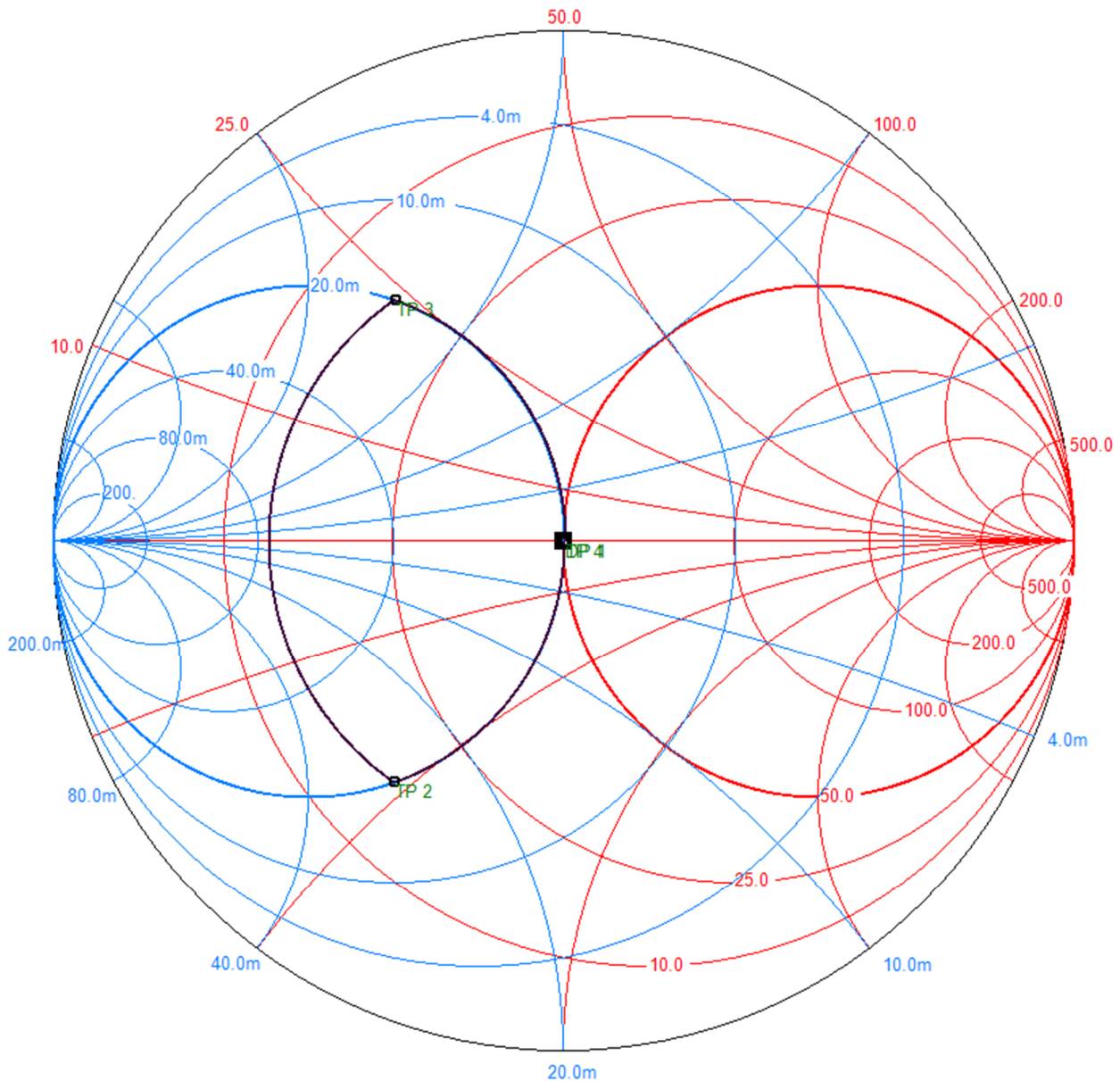
$$\frac{l_s}{\lambda} = 0,06436 \rightarrow \underline{\underline{l_s = 0,06436 \cdot 141,4\text{mm} = 9,1\text{mm}}} \quad \left[\underline{Z} = j20\Omega : \frac{l_s}{\lambda} = 0,0608 \quad l_s = 8,6\text{mm} \right]$$

3. Ohne L ist Leitung 1 mit Z_L abgeschlossen $\rightarrow 50\Omega$ parallel zu C :

$$\underline{Z}'_1 = \frac{R \cdot jX_C}{R + jX_C} = \frac{50\Omega \cdot (-j35,37\Omega)}{50\Omega - j35,37\Omega} = (16,676 - j23,57)\Omega;$$

$$\text{normiert: } \frac{\underline{Z}'_1}{Z_L} = \frac{(16,676 - j23,57)\Omega}{50\Omega} = 0,3335 - j0,4715$$

$\rightarrow$ ins Smith-Chart eintragen oder: Einzelwerte R und C in Smith-Chart eintragen.



$$\frac{l'_2}{\lambda} = 0,07615 - 0,42385 + 0,5 = 0,1523 \quad \underline{\underline{l'_2 = 0,1523 \cdot 141,4\text{mm} = 21,5\text{mm}}}$$

$$B_{C2} \cdot Z_L = 1,419 \rightarrow \omega C_2 = \frac{1,419}{Z_L} \quad \underline{\underline{C_2 = \frac{1,419}{\omega \cdot Z_L} = \frac{1,419}{2\pi \cdot 1,5 \cdot 10^9 \frac{1}{s} \cdot 50 \frac{V}{A}} = 3,0\text{pF}}}$$

Aufgabe 2:

(21 Punkte)

$$1. \quad \left(\frac{s}{h}\right)_{\min} = \frac{125\mu\text{m}}{0,5\text{mm}} = \frac{125\mu\text{m}}{500\mu\text{m}} = 0,25 \quad \text{aus 7-12: } \frac{s}{h} = 0,25 \rightarrow k_L \approx 0,4$$

$$\rightarrow \underline{s_{41,\max}} = k_L = 0,4 \quad \underline{C_{k,\min}} = 20\text{dB} \cdot \lg\left(\frac{1}{0,4}\right) = \underline{8,0\text{dB}} \quad (\text{oder aus Diagramm Seite 7-14})$$

2.

$$Z_L = 50\Omega; \quad \varepsilon_r = 4,5 \rightarrow \text{aus Diagramm: } \frac{w}{h} = 2,0 \quad \text{damit (aus Diagramm): } \varepsilon_{r,\text{eff}} \approx 3,4$$

$$\lambda = \frac{c_0}{\sqrt{\varepsilon_{r,\text{eff}} \cdot f_0}} = \frac{3 \cdot 10^8 \frac{\text{m}}{\text{s}}}{\sqrt{3,4 \cdot 2,0 \cdot 10^9 \frac{1}{\text{s}}}} = 81,35\text{mm}$$

$$\text{minimale Koppeldämpfung für } \underline{l} = \frac{\lambda}{4} = \frac{81,35\text{mm}}{4} = \underline{20,3\text{mm}}$$

3.

$$\text{bei } f = 3,0\text{GHz: } \beta \cdot l = \beta_0 \cdot \frac{3,0\text{GHz}}{2,0\text{GHz}} \cdot l = \frac{3}{2} \cdot \beta_0 \cdot l = \frac{3}{2} \cdot 90^\circ = 135^\circ$$

$$\sin 135^\circ = \frac{1}{2}\sqrt{2} \quad \cos 135^\circ = \frac{1}{2}\sqrt{2} \quad k_L^2 = \left(10^{\frac{13\text{dB}}{20\text{dB}}}\right)^2 = 0,05012$$

$$|s_{41,\min}|^2 = \frac{k_L^2 \cdot \sin^2 135^\circ}{(1 - k_L^2) \cdot \cos^2 135^\circ + \sin^2 135^\circ} = \frac{0,05012 \cdot \frac{1}{2}}{(1 - 0,05012) \cdot \frac{1}{2} + \frac{1}{2}} = 0,0257$$

$$\rightarrow \underline{C_{k,\max}} = 10\text{dB} \cdot \lg \frac{1}{|s_{41,\min}|^2} = 10\text{dB} \cdot \lg \frac{1}{0,0257} = \underline{15,9\text{dB}}$$

4.

$$\left(\frac{w}{h}\right)_{\min} = \frac{125\mu\text{m}}{500\mu\text{m}} = 0,25 \rightarrow \text{aus Diagramm: } Z_{L,\max} \approx 120\Omega$$

$$Z_{L2} = Z_L \cdot \frac{\sqrt{1 - k^2}}{k}; \quad \left(\frac{Z_{L2}}{Z_L}\right)^2 = \frac{1 - k^2}{k^2}; \quad k^2 \cdot \left[\left(\frac{Z_{L2}}{Z_L}\right)^2 + 1\right] = 1; \quad k^2 = \frac{1}{\left(\frac{Z_{L2}}{Z_L}\right)^2 + 1}$$

$$k_{\min}^2 = \frac{1}{\left(\frac{Z_{L,\max}}{Z_L}\right)^2 + 1} = \frac{1}{\left(\frac{120\Omega}{50\Omega}\right)^2 + 1} = 0,1479$$

$$\underline{C_{k,\max}} = 10\text{dB} \cdot \lg \frac{1}{k_{\min}^2} = 10\text{dB} \cdot \lg \frac{1}{0,1479} = \underline{8,3\text{dB}}$$

Aufgabe 3:

(15 Punkte)

1.

$$\underline{Z}_E = Z_L \cdot \frac{1 + \underline{s}_{11}}{1 - \underline{s}_{11}}; \quad 1 + \underline{s}_{11} = (1 - \underline{s}_{11}) \cdot \frac{\underline{Z}_E}{Z_L} = \frac{\underline{Z}_E}{Z_L} - \frac{\underline{Z}_E}{Z_L} \cdot \underline{s}_{11}; \quad \left(1 + \frac{\underline{Z}_E}{Z_L}\right) \cdot \underline{s}_{11} = \frac{\underline{Z}_E}{Z_L} - 1; \quad \underline{s}_{11} = \frac{\frac{\underline{Z}_E}{Z_L} - 1}{\frac{\underline{Z}_E}{Z_L} + 1}$$

$$\underline{s}_{11} = \frac{\frac{(10 + j20)\Omega}{50\Omega} - 1}{\frac{(10 + j20)\Omega}{50\Omega} + 1} = 0,7071 \cdot e^{j135^\circ} \quad \text{analog:} \quad \underline{s}_{22} = \frac{\frac{\underline{Z}_A}{Z_L} - 1}{\frac{\underline{Z}_A}{Z_L} + 1} = \frac{\frac{(5 - j10)\Omega}{50\Omega} - 1}{\frac{(5 - j10)\Omega}{50\Omega} + 1} = 0,8246 \cdot e^{-j157^\circ}$$

2.

$$\underline{v}_{Tr,max} = 10\text{dB} \cdot \lg \left(|\underline{s}_{21}|^2 \cdot \frac{1}{1 - |\underline{s}_{11}|^2} \cdot \frac{1}{1 - |\underline{s}_{22}|^2} \right) = 20,0\text{dB}; \quad |\underline{s}_{21}|^2 \cdot \frac{1}{1 - |\underline{s}_{11}|^2} \cdot \frac{1}{1 - |\underline{s}_{22}|^2} = 10^{\frac{20\text{dB}}{10\text{dB}}} = 100$$

$$|\underline{s}_{21}|^2 = 100 \cdot (1 - |\underline{s}_{11}|^2) \cdot (1 - |\underline{s}_{22}|^2) = 100 \cdot (1 - 0,7071^2) \cdot (1 - 0,8246^2) = 16,0$$

$$\rightarrow \underline{|\underline{s}_{21}| = 4,0}$$

3.

$$\begin{aligned} |\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}|_{min} &= |0,7071 \cdot 0,8246 \cdot e^{j(\varphi_{11} + \varphi_{22})} - 0,1 \cdot 4,0 \cdot e^{j(\varphi_{12} + \varphi_{21})}| = \\ &= |(0,7071 \cdot 0,8246 - 0,1 \cdot 4,0) \cdot e^{j(\varphi_{12} + \varphi_{21})}| = \\ &= |0,7071 \cdot 0,8246 - 0,1 \cdot 4,0| = 0,1831 \end{aligned}$$

$$K = \frac{1 - |\underline{s}_{11}|^2 - |\underline{s}_{22}|^2 + |\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}|_{min}^2}{2|\underline{s}_{12}\underline{s}_{21}|} = \frac{1 - 0,7071^2 - 0,8246^2 + 0,1831^2}{2 \cdot 0,1 \cdot 4,0} = -0,183 < 1$$

Transistor ist **nicht** absolut stabil!

Aufgabe 4:

(17 Punkte)

1.

$$G_1 = 10^{\frac{26\text{dB}}{10\text{dB}}} = 400 \quad G_2 = 10^{\frac{6\text{dB}}{10\text{dB}}} = 4,0 \quad G_L = 10^{\frac{-10\text{dB}}{10\text{dB}}} = 0,1$$

$$F_1 = 10^{\frac{6\text{dB}}{10\text{dB}}} = 4,0 \quad F_2 = 10^{\frac{2\text{dB}}{10\text{dB}}} = 1,585 \quad F_L = \frac{1}{G_L} = 10$$

höchste Rauschzahl:

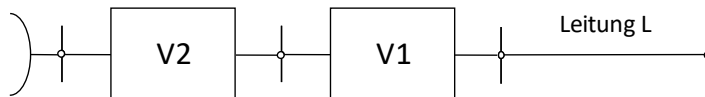
Antenne



$$\underline{\underline{F_{ges,max}}} = 1 + (F_L - 1) + \frac{F_1 - 1}{G_L} + \frac{F_2 - 1}{G_L \cdot G_1} = 1 + (10 - 1) + \frac{4,0 - 1}{0,1} + \frac{1,585 - 1}{0,1 \cdot 400} = 40,01 \hat{=} \underline{\underline{16,0\text{dB}}}$$

niedrigste Rauschzahl:

Antenne



$$\underline{\underline{F_{ges,min}}} = 1 + (F_2 - 1) + \frac{F_1 - 1}{G_2} + \frac{F_L - 1}{G_2 \cdot G_1} = 1 + (1,585 - 1) + \frac{4,0 - 1}{4,0} + \frac{10 - 1}{4,0 \cdot 400} = 2,34 \hat{=} \underline{\underline{3,7\text{dB}}}$$

2.

$$P_{R,ein} = k \cdot T_A \cdot B = 1,38 \cdot 10^{-23} \frac{\text{Ws}}{\text{K}} \cdot 290\text{K} \cdot 20 \cdot 10^6 \frac{1}{\text{s}} = 8,0 \cdot 10^{-14} \text{ W}$$

$$P_{S,ein} = 1,0\text{mW} \cdot 10^{\frac{-80\text{dBm}}{10\text{dB}}} = 1,0 \cdot 10^{-11} \text{ W}$$

$$\left(\frac{S}{N} \right)_{ein} = \frac{P_{S,ein}}{P_{R,ein}} = \frac{1,0 \cdot 10^{-11} \text{ W}}{8,0 \cdot 10^{-14} \text{ W}} = 125 \hat{=} 21,0\text{dB}$$

$$\left(\frac{S}{N} \right)_{ein} = F_{ges} \cdot \left(\frac{S}{N} \right)_{aus} \quad \left(\frac{S}{N} \right)_{aus} = \frac{\left(\frac{S}{N} \right)_{ein}}{10^{\frac{10\text{dB}}{10\text{dB}}}} = \frac{125}{10} = 12,5 \hat{=} \underline{\underline{11,0\text{dB}}}$$

$$\text{oder: } \left(\frac{S}{N} \right)_{aus} \hat{=} 21,0\text{dB} - 10\text{dB} = 11,0\text{dB}$$

Aufgabe 5:

(16 Punkte)

1. Reflexionsfaktor der Ersatzlast:

$$\underline{r}_1 = \underline{s}_{11} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_L}{1 - \underline{s}_{22} \cdot \underline{r}_L} = j0,4 + \frac{0,7 \cdot 0,7 \cdot (0,5 + j0,5)}{1 - j0,4 \cdot (0,5 + j0,5)} = (0,166 + j0,632) = 0,653 \cdot e^{j75,3^\circ}$$

für maximale Leistung: Anpassung

$$\underline{r}_Q = \underline{r}_1^* = (0,166 - j0,632) = 0,653 \cdot e^{-j75,3^\circ}$$

2. entspricht der Leistung, die die Quelle bei Anpassung abgeben kann:

$$\underline{\underline{P_{Q,max}}} = \frac{1}{2} |\underline{b_{Q,max}}|^2 = \frac{1}{2} |\underline{b_Q}|^2 \cdot \frac{1}{1 - |\underline{r_Q}|^2} = 5,0 \text{ W} \cdot \frac{1}{1 - 0,653^2} = \underline{\underline{8,7 \text{ W}}}$$

3. Ersatzquelle:

$$\underline{r}_2 = \underline{s}_{22} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_Q}{1 - \underline{s}_{11} \cdot \underline{r}_Q} = j0,4 + \frac{0,7 \cdot 0,7 \cdot (0,169 - j0,631)}{1 - j0,4 \cdot (0,169 - j0,631)} = 0,147 - j0,00007 \approx 0,147$$

$$\underline{b_{Q2}} = \underline{b_Q} \cdot \frac{\underline{s}_{21}}{1 - \underline{s}_{11} \cdot \underline{r}_Q} = \underline{b_Q} \cdot \frac{0,7}{1 - j0,4 \cdot (0,169 - j0,631)} = \underline{b_Q} \cdot (0,929 + j0,0084) \approx \underline{b_Q} \cdot 0,929$$

mit 4-18:

$$\underline{\underline{P_L}} = \frac{1}{2} |\underline{b_{Q2}}|^2 \cdot \frac{1 - |\underline{r}_L|^2}{|1 - \underline{r}_2 \cdot \underline{r}_L|^2} = 5,0 \text{ W} \cdot (0,929)^2 \cdot \frac{1 - |0,5 + j0,5|^2}{|1 - 0,147 \cdot (0,5 + j0,5)|^2} = 4,32 \text{ W} \cdot 0,579 = \underline{\underline{2,50 \text{ W}}}$$